

Initial Allocation Model

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Risk Free Asset: Cox–Ingersoll–Ross (CIR) Model

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t$$

- r_t : The short-term interest rate at time t .
- κ : Speed of mean reversion — how quickly r_t moves toward the long-run mean.
- θ : Long-run mean level of the short term rate.
- σ : Volatility parameter; higher σ means more random fluctuations. It is strictly positive.
- $\sqrt{r_t}$: State-dependent volatility; ensures non-negative rates.
- dZ_t : Represents the stochastic shock over the infinitesimal interval $[t, t + dt]$.

Sign Analysis of Terms in the CIR Model

Term	Description	Possible Values
r_t	Short-term interest rate	≥ 0
θ	Mean short-term rate	> 0
κ	Mean reversion speed	> 0
$\theta - r_t$	Distance from long-run mean	$> 0, = 0, \text{ or } < 0$
dt	Infinitesimal time increment	> 0
$\kappa(\theta - r_t)$	Deterministic mean-reversion force	$> 0, = 0, \text{ or } < 0$
$\kappa(\theta - r_t)dt$	Drift term	$> 0, = 0, \text{ or } < 0$
$\sqrt{r_t}$	Volatility scale (CIR-specific)	≥ 0 , zero when $r_t = 0$
σ	Volatility of volatility	> 0
dZ_t	Standard Brownian motion increment	$> 0, = 0, \text{ or } < 0$
$\sigma\sqrt{r_t}$	Volatility coefficient	≥ 0
$\sigma\sqrt{r_t}dZ_t$	Diffusion term	$> 0, = 0, \text{ or } < 0$
dr_t	Interest rate differential	$> 0, = 0, \text{ or } < 0$

State-dependent volatility

The randomness is proportional to $\sqrt{r_t}$, which **grows when r_t is large**, and **shrinks when r_t is small**. This is more realistic than constant volatility because **interest rates don't fluctuate wildly when they're near zero**.

Keeps $r_t \geq 0$ (non-negative interest rates)

A natural question is: what prevents r_t from becoming negative, especially since the square root $\sqrt{r_t}$ becomes undefined for $r_t < 0$? The answer lies in the structure of the diffusion and drift terms. As $r_t \rightarrow 0$, the volatility $\sigma\sqrt{r_t}$ also tends to zero, meaning that random fluctuations vanish near the boundary. At the same time, the drift term becomes approximately $\kappa\theta dt$, which is strictly positive and pushes the process upward. Thus, the process becomes increasingly deterministic and mean-reverting as it approaches zero. This dynamic causes the process to “bounce off” the boundary rather than cross it. A formal guarantee of non-negativity is provided by the *Feller condition*, which states that if $2\kappa\theta \geq \sigma^2$, then the boundary at $r_t = 0$ is inaccessible and the process remains strictly non-negative almost surely. If the Feller condition is violated, the process may still stay non-negative, but can touch zero; in severe cases, it may cross below zero, at which point the square root becomes undefined and the SDE breaks down. Therefore, both the vanishing of the volatility near zero and the strength of the mean-reverting drift explain why the CIR model is able to maintain $r_t \geq 0$.

Dimensional Analysis of the CIR Model

Consider the Cox–Ingersoll–Ross (CIR) model for the short-term interest rate:

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t$$

What Are the Units?

At first glance, one might think that r_t , being a “rate,” should have units of $\frac{1}{\text{time}}$, and that κ is a unitless scalar. However, this is not the case in finance. Let’s break it down carefully.

Units of r_t

The interest rate r_t is a **dimensionless scalar** — it represents a proportion (e.g., 5% interest means $r_t = 0.05$).

$$[r_t] = 1$$

Left-Hand Side — dr_t

Since r_t is dimensionless, its infinitesimal change dr_t must also be dimensionless:

$$[dr_t] = 1$$

Drift Term — $\kappa(\theta - r_t) dt$

We now examine the units of each piece:

- $\theta - r_t$: dimensionless
- dt : units of time

To ensure dimensional consistency with dr_t , we need:

$$[\kappa] = \frac{1}{\text{time}} \quad \Rightarrow \quad [\kappa(\theta - r_t)dt] = \frac{1}{\text{time}} \cdot 1 \cdot \text{time} = 1$$

So the drift term is dimensionless, just like dr_t .

Diffusion Term — $\sigma\sqrt{r_t} dZ_t$

Now for the stochastic component:

- $\sqrt{r_t}$: still dimensionless, since r_t is dimensionless
- dZ_t : Brownian motion increment, which has units of $\sqrt{\text{time}}$

To cancel the time units and ensure this term is dimensionless, we require:

$$[\sigma] = \frac{1}{\sqrt{\text{time}}} \Rightarrow [\sigma\sqrt{r_t} dZ_t] = \frac{1}{\sqrt{\text{time}}} \cdot 1 \cdot \sqrt{\text{time}} = 1$$

Risky Asset: Heston Stochastic Volatility Model Equation One

$$dP_1(t) = \mu P_1(t) dt + \sqrt{v_t} P_1(t) dZ_t$$

- $P_1(t)$: Price of the risky asset at time t .
- μ : Expected return on the asset (1 / unit of time).
- $\sqrt{v_t}$: Stochastic volatility — depends on variance process v_t .
- dZ_t : Represents the stochastic shock over the infinitesimal interval $[t, t + dt]$.

Units of Brownian Motion W_t

By definition:

$$\text{Var}(Z_t) = \mathbb{E}[(Z_t - \mathbb{E}[Z_t])^2]$$

For a standard Brownian motion, we know:

$$\mathbb{E}[Z_t] = 0 \Rightarrow \text{Var}(Z_t) = \mathbb{E}[Z_t^2]$$

Also, from Brownian motion properties:

$$Z_t \sim \mathcal{N}(0, t) \Rightarrow \text{Var}(Z_t) = t$$

So putting it together:

$$\mathbb{E}[Z_t^2] = t$$

Assume time t is measured in seconds. Then:

$$[\mathbb{E}[Z_t^2]] = \text{seconds} \Rightarrow [Z_t^2] = \text{seconds} \Rightarrow [Z_t] = \sqrt{\text{seconds}}$$

In the Heston stochastic differential equation:

$$dP_1(t) = \mu P_1(t) dt + \sqrt{v_t} P_1(t) dZ_t$$

We require both terms on the right-hand side to have units of dollars.

Drift term

$$[\mu] = 1/\text{time}, \quad [dt] = \text{time} \Rightarrow [\mu P_1(t) dt] = \text{dollars}$$

Diffusion term

$$[v_t] = 1/\text{time} \Rightarrow [\sqrt{v_t}] = 1/\sqrt{\text{time}}$$

$$[dZ_t] = \sqrt{\text{time}}$$

$$\text{So } [\sqrt{v_t} \cdot dZ_t] = 1 \Rightarrow [\sqrt{v_t} P_1(t) dZ_t] = \text{dollars}$$

Risky Asset: Heston Stochastic Volatility Model Equation Two

$$dv_t = \kappa(\theta - v_t) dt + \sigma\sqrt{v_t} dZ_t$$

- v_t : Stochastic variance (i.e., $v_t = \sigma_t^2$).
- κ : Speed of mean reversion for variance.
- θ : Long-run mean of variance.
- σ : Volatility of variance — controls how volatile v_t is.
- dZ_t : Represents the stochastic shock over the infinitesimal interval $[t, t + dt]$.
- $\sqrt{v_t}$: Ensures variance stays non-negative.

Dimensional Analysis of the Heston Variance Process

Consider the stochastic variance equation in the Heston model:

$$dv_t = \kappa(\theta - v_t) dt + \sigma\sqrt{v_t} dZ_t$$

Units of v_t

The process v_t represents the *instantaneous variance* of the asset return process, meaning:

$$v_t = \sigma_t^2 \Rightarrow [v_t] = \left(\frac{1}{\sqrt{\text{time}}} \right)^2 = \frac{1}{\text{time}}$$

Hence:

$$[dv_t] = \frac{1}{\text{time}}$$

Drift Term: $\kappa(\theta - v_t) dt$

- θ and v_t both have units of 1/time
- $\theta - v_t$: has units of 1/time
- dt : has units of time

To make the product dimensionally consistent with dv_t , we require:

$$[\kappa] = \frac{1}{\text{time}}$$

Then:

$$[\kappa(\theta - v_t)dt] = \frac{1}{\text{time}} \cdot \frac{1}{\text{time}} \cdot \text{time} = \frac{1}{\text{time}}$$

Diffusion Term: $\sigma\sqrt{v_t} dZ_t$

- $\sqrt{v_t}$: has units $1/\sqrt{\text{time}}$
- dZ_t : has units $\sqrt{\text{time}}$

To ensure the entire term has units 1/time, we require:

$$[\sigma] = \frac{1}{\text{time}}$$

Then:

$$[\sigma\sqrt{v_t} dZ_t] = \frac{1}{\text{time}} \cdot \frac{1}{\sqrt{\text{time}}} \cdot \sqrt{\text{time}} = \frac{1}{\text{time}}$$

Wealth Process with Consumption Term

$$dW^S(t) = S(t) \cdot \frac{dP_1(t)}{P_1(t)} + (W(t) - S(t)) \cdot \frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t) dt - C(t) dt$$

- $S(t) \frac{dP_1(t)}{P_1(t)}$: Return from wealth invested in the risky asset.
- $(W(t) - S(t)) \frac{dP_0(t)}{P_0(t)}$: Return from the wealth invested in the risk-free asset.
- $-(\vartheta + \beta - d)S(t) dt$: Adjustment for transaction costs, taxes, and dividends.
- $-C(t) dt$: Rate of consumption, which directly reduces wealth over time.
- $C(t)$: Control variable representing the investor's consumption rate at time t .

Note: We Assume consumption is a fixed proportion of wealth:

$$C(t) = \kappa W(t), \quad \kappa \in (0, 1)$$

Plugging in CIR and Heston into the Wealth Equation

First, we substitute the Heston model for the risky asset

$$\frac{dP_1(t)}{P_1(t)} = \mu dt + \sqrt{v_t} dZ_t.$$

Next, because the price process of the risk-free asset satisfies the ordinary differential equation,

$$dP_0(t) = r_t P_0(t) dt,$$

we substitute

$$\frac{dP_0(t)}{P_0(t)} = r_t dt$$

for the risk-free asset.

Note: Here, the process r_t is used symbolically to define the return on the risk-free asset. We do not need a closed-form solution for r_t ; it is treated as a state variable whose evolution is governed by the CIR SDE.

Substituted Wealth Process with Consumption Term

$$\begin{aligned} dW^S(t) &= S(t) \cdot (\mu dt + \sqrt{v_t} dZ_t) + (W(t) - S(t)) \cdot r_t dt - (\vartheta + \beta - d)S(t) dt - C(t) dt \\ &= \mu S(t) dt + \sqrt{v_t} S(t) dZ_t + W(t) r_t dt - S(t) r_t dt - (\vartheta + \beta - d)S(t) dt - C(t) dt \\ &= [\mu S(t) + W(t) r_t - S(t) r_t - (\vartheta + \beta - d)S(t) - C(t)] dt + \sqrt{v_t} S(t) dZ_t \\ &= [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)] dt + \sqrt{v_t} S(t) dZ_t \end{aligned}$$

Deriving the Hamilton-Jacobi-Bellman Equation

We define the value function $J(t, r(t), W(t))$, where:

- t is time,
- $r(t)$ is the short rate process,
- $W(t)$ is the wealth process,
- $S(t)$ is the control (investment in risky asset).

Itô's Lemma (Multivariate Form)

Let $J(t, r(t), W(t))$, where both $r(t)$ and $W(t)$ are Itô processes of the form $dX_t = \mu_t dt + \sigma_t dZ(t)$, driven by the same Brownian motion $Z(t)$. Then:

$$dJ = J_t dt + J_r dr + J_w dW + \frac{1}{2} J_{rr} (dr)^2 + \frac{1}{2} J_{ww} (dW)^2 + J_{rw} dr dW$$

Computing Quadratic Variation Terms

We apply Itô's Lemma to the value function $J(t, r_t, W_t)$, where both r_t and W_t are Itô processes driven by the same Brownian motion Z_t .

We use the fundamental stochastic calculus identities:

$$(dZ_t)^2 = dt, \quad dt \cdot dZ_t = 0, \quad (dt)^2 = 0$$

Short Rate Process

Recall the Short Rate Process (Cox–Ingersoll–Ross):

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t.$$

We compute the second-order differential $(dr_t)^2$ as follows:

$$\begin{aligned} (dr_t)^2 &= [\kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t]^2 \\ &= \kappa^2(\theta - r_t)^2 (dt)^2 + 2\kappa(\theta - r_t)\sigma\sqrt{r_t} dt dZ_t + \sigma^2 r_t (dZ_t)^2 \\ &= 0 + 0 + \sigma^2 r_t dt \\ &= \sigma^2 r_t dt \end{aligned}$$

Wealth Process

The wealth process is given by:

$$dW^S(t) = [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)] dt + \sqrt{v_t} S(t) dZ_t$$

Define:

$$A(t) := \mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)$$

Then the wealth process becomes:

$$dW^S(t) = A(t) dt + \sqrt{v_t} S(t) dZ_t$$

Now compute the square of the differential:

$$\begin{aligned} (dW^S(t))^2 &= [A(t) dt + \sqrt{v_t} S(t) dZ_t]^2 \\ &= A(t)^2 (dt)^2 + 2A(t)\sqrt{v_t} S(t) dt dZ_t + v_t S(t)^2 (dZ_t)^2 \\ &= 0 + 0 + v_t S(t)^2 dt \\ &= v_t S(t)^2 dt \end{aligned}$$

Cross Term

We compute the cross-variation term between the short rate process and the wealth process:

$$\begin{aligned}
dr_t \cdot dW^S(t) &= [\kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t] \cdot [A(t) dt + \sqrt{v_t}S(t) dZ_t] \\
&= \kappa(\theta - r_t)A(t) (dt)^2 + \kappa(\theta - r_t)\sqrt{v_t}S(t) dt dZ_t + \sigma\sqrt{r_t}A(t) dZ_t dt + \sigma\sqrt{r_t}\sqrt{v_t}S(t) (dZ_t)^2 \\
&= 0 + 0 + 0 + \sigma\sqrt{r_tv_t}S(t) dt \\
&= \sigma\sqrt{r_tv_t}S(t) dt
\end{aligned}$$

These second-order terms are required when applying Itô's Lemma to $J(t, r_t, W_t)$, and appear in the final HJB equation through the terms:

$$\frac{1}{2}J_{rr}(dr_t)^2, \quad \frac{1}{2}J_{ww}(dW_t)^2, \quad J_{rw} dr_t \cdot dW_t$$

Applying Itô's Lemma (Multivariate Form)

$$\begin{aligned}
dJ &= J_t dt + J_r [\kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dZ_t] \\
&\quad + J_w [(\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)) dt + \sqrt{v_t}S(t) dZ_t] \\
&\quad + \frac{1}{2}J_{rr} \sigma^2 r_t dt + \frac{1}{2}J_{ww} v_t S(t)^2 dt + J_{rw} \sigma\sqrt{r_tv_t} S(t) dt
\end{aligned}$$

Note: Within this context, $dr_t = d_r$.

Extract Drift Terms for the HJB

The HJB uses only drift terms (the dt -terms). Group them:

$$\begin{aligned}
0 &= J_t + \kappa(\theta - r_t) J_r \\
&\quad + [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)] J_w \\
&\quad + \frac{1}{2}\sigma^2 r_t J_{rr} + \frac{1}{2}v_t S(t)^2 J_{ww} + \sigma\sqrt{r_tv_t} S(t) J_{rw}
\end{aligned}$$

This is the Hamilton-Jacobi-Bellman (HJB) equation corresponding to the wealth and short rate dynamics under control $S(t)$.

Finding the Optimal Control

Taking the derivative with respect to $S(t)$, we get:

- $\mu S(t)J_w \Rightarrow \mu J_w$
- $-r_t S(t)J_w \Rightarrow -r_t J_w$
- $-(\vartheta + \beta - d)S(t)J_w \Rightarrow -(\vartheta + \beta - d)J_w$
- $\frac{1}{2}v_t S(t)^2 J_{ww} \Rightarrow v_t S(t)J_{ww}$
- $\sigma\sqrt{r_tv_t}S(t)J_{rw} \Rightarrow \sigma\sqrt{r_tv_t}J_{rw}$

Combining all terms:

$$\frac{dF}{dS(t)} = (\mu - r_t - \vartheta - \beta + d)J_w + v_t S(t)J_{ww} + \sigma\sqrt{r_tv_t}J_{rw}$$

$$\text{Set } \frac{dF}{dS(t)} = 0 : \quad [\mu + d - r_t - (\vartheta + \beta)] J_w + \sigma\sqrt{r_tv_t}J_{rw} + v_t S(t)J_{ww} = 0$$

Solve for the optimal control $S^*(t)$:

$$S^*(t) = -\frac{[(\mu + d) - (r_t + \vartheta + \beta)]J_w + \sigma\sqrt{r_tv_t}J_{rw}}{v_t J_{ww}} \quad (34)$$

Applying the CRRA Ansatz to Eliminate Wealth Dependency

Eliminating Dependency on w

To eliminate dependency on w , we conjecture that

$$J(t, r, w) = G(t, r) \cdot \frac{w^{1-\phi}}{1-\phi}, \quad \phi \neq 1 \quad (35a)$$

with the boundary condition

$$G(T, r) = 1.$$

Note: Since $r(t) = r$ and $W(t) = w$, we can write:

$$J(t, r(t), W(t)) = J(t, r, w)$$

From (35a), we obtain:

$$J_w = w^{-\phi} G, \quad J_{ww} = -\phi w^{-\phi-1} G, \quad \text{and} \quad J_{rw} = w^{-\phi} G_r \quad (35c)$$

Substituting (35c) into the Optimal Control Formula

We begin with the optimal control expression from the new equation:

$$S^*(t) = - \frac{[(\mu + d) - (r_t + \vartheta + \beta)] J_w + \sigma \sqrt{r_t v_t} J_{rw}}{v_t J_{ww}}$$

From equation (35c), we know the following expressions for the derivatives of $J(t, r, w)$:

$$\begin{aligned} J_w &= w^{-\phi} G, \\ J_{ww} &= -\phi w^{-\phi-1} G, \\ J_{rw} &= w^{-\phi} G_r \end{aligned}$$

Substitute these into the expression for $S^*(t)$:

$$\begin{aligned} S^*(t) &= - \frac{[(\mu + d) - (r_t + \vartheta + \beta)] \cdot w^{-\phi} G + \sigma \sqrt{r_t v_t} \cdot w^{-\phi} G_r}{v_t \cdot (-\phi w^{-\phi-1} G)} \\ &= \frac{[(\mu + d) - (r_t + \vartheta + \beta)] \cdot w^{-\phi} G + \sigma \sqrt{r_t v_t} \cdot w^{-\phi} G_r}{\phi v_t w^{-\phi-1} G} \end{aligned}$$

Now simplify numerator and denominator:

$$S^*(t) = \frac{w}{\phi v_t} \left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma \sqrt{r_t v_t} \cdot \frac{G_r}{G} \right] \quad (36)$$

This is the final closed-form expression for the optimal strategy $S^*(t)$ in terms of current wealth w , short rate r_t , and the reduced value function $G(t, r)$.

Eliminating Dependency on r

Further to eliminate dependency on r we conjecture that

$$G(t, r) = h(t) \left[\frac{r^{1-\phi}}{1-\phi} \right], \quad (37a)$$

such that

$$h(T) = \frac{1-\phi}{r^{1-\phi}}. \quad (37b)$$

We obtain from (37a)

$$G_r = r^{-\phi} h(t). \quad (37c)$$

Substituting (37a) and (37c) into (36)

From equation (36), we have:

$$S^*(t) = \frac{w}{\phi v_t} \left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma \sqrt{r_t v_t} \cdot \frac{G_r}{G} \right]. \quad (36)$$

Using the ansatz from (37a)

$$G(t, r) = h(t) \left[\frac{r^{1-\phi}}{1-\phi} \right], \quad (37a)$$

and its derivative,

$$G_r = r^{-\phi} h(t), \quad (37c)$$

we compute

$$\frac{G_r}{G} = \frac{r^{-\phi} h(t)}{h(t) \cdot \frac{r^{1-\phi}}{1-\phi}} = \frac{(1-\phi)r^{-\phi}}{r^{1-\phi}} = \frac{(1-\phi)}{r}.$$

Substitute this into (36):

$$\begin{aligned} S^*(t) &= \frac{w}{\phi v_t} \left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma \sqrt{r_t v_t} \cdot \frac{1-\phi}{r_t} \right] \\ &= \frac{w}{\phi v_t} \left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma \sqrt{v_t} \cdot \frac{1-\phi}{\sqrt{r_t}} \right]. \end{aligned}$$

This expression eliminates dependency on G , leaving $S^*(t)$ in terms of w (the wealth of the investor at time t), v_t , and r_t only.

Note: To evaluate the expression for $S^*(t)$ in practice, we can substitute the long-run average values of the state variables. Specifically, we replace:

$$r_t \approx \theta_r, \quad v_t \approx \theta_v$$

where θ_r is the long-run mean of the short rate (from the CIR process) and θ_v is the long-run mean of the variance process (from the Heston model). This approximation simplifies the control expression while preserving its qualitative behavior.

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